

top_module.v

```
module top_module (  
    input clk,  
    input rst,  
    input activity_in,  
    output [3:0] out  
);  
wire enable;  
wire [1:0] state;  
energy_aware_controller ctrl (  
    .clk(clk),  
    .rst(rst),  
    .activity_in(activity_in),  
    .enable(enable),  
    .state(state)  
);  
functional_block fb (  
    .clk(clk),  
    .rst(rst),  
    .enable(enable),  
    .counter(out)  
);  
endmodule
```